

2.11 PRACTICAL SESSIONS

Sessionwise practical problems are given as following:

Session-1

1. Download and install WEKA. Navigate the various options available in WEKA. Explore the available datasets in WEKA. Load various datasets and observe the following”
 - a. List the attribute names and their types
 - b. No. of records in each dataset
 - c. Identify the class attribute(if any)
 - d. Plot Histogram
 - e. Determine the no. of records for each class.
 - f. Visualize the data in different dimensions.
2. Create your own EXCEL file. Convert the EXCEL file to .csv format and prepare it as .arff file.
3. Try to create your own datasets.
4. Preprocess and classify Customer, Argiculture, Weather, Whole-sale Customers or the datasets of your own choice from <https://archive.ics.uci.edu/ml/datasets.php>

Session-2

5. Perform the basic pre-processing operations on data relation such as removing an attribute and filter attribute bank data
6. Demonstrate the preprocessing mechanism on the following datasets:
 - a. student.arff
 - b. labor.arff
 - c. contactlenses.arff

Session-3

7. Perform the following:
 - Explore various options available in WEKA for preprocessing data and apply unsupervised filters like Discretization, Resample-filter etc.. on various datasets.
 - Load weather, nominal, Iris, Glass datasets into WEKA and run Apriori algorithms with different support and confidence values.
 - Study the rules generated.
 - Apply different discretization filters on numerical attributes and run the Apriori association rule algorithm. Study the rules generated.

- Derive interesting insights and observe the effect of discretization in the rule generation process.
8. Implement the Apriori Algorithm to find the association rules in contactless.arff dataset.

Session-4

9. Find the frequent patterns using FP-Growth algorithm on contactlenses.arff and test.arff datasets.
10. Generate association rules using Apriori algorithm with Bank.arff relation
 - a. Set minimum support range as 20% -100% incremental decrease factor as 5% and confidence factor as 80% and generate 5 rules.
 - b. Set minimum support as 10%, delta 5%, minimum average(4ft) as 150% and generate 4 rules.
11. Generate association rule for the credit card promotion dataset using a priori algorithm with the support range 40% to 100% confidence as 10% incremental decrease as 5% and generate 6 rules.

Session-5

12. Perform the following:
 - Use contactlenses.arff and load it into WEKA. Check that all attributes are nominal (categorical).
 - Change to the Associate Panel. Select “Apriori” as associator. After pressing the start button, Apriori starts to build its model and writes its output into the output field. The first part of the output (“Run information”) describes the options that have been set and the data set used. Make sure you understand all the data reported.
 - The rules that have been generated are listed at the end of the output. By default, only the 10 most valuable rules according to their confidence level are shown. Each rule consists of some attribute values on a left hand side of the arrow, the arrow sign and the right hand side list of attribute values. Right of the arrow sign are the predicted attribute values. Rules have certain support and confidence values. The number before the arrow sign is the number of instances the rule applies to. The number after the arrow sign is the number of instances predicted correctly. The number in brackets after ‘conf:’ is the confidence of the rule. Analyse the rules mined from the data set. What are their confidence and support values? Examine the number of large itemsets – make sure you understand how this data has been calculated (check that the values you would get ‘manually’ are correct).

Session-6

13. Perform the following:
 - Use zoo.arff dataset and load it into WEKA. Examine the attributes and make sure you understand their meaning. Are all attributes nominal?

- In the preprocess area, deselect the animal and legs attributes. The animal attribute is the name of the animal, and is not useful for mining. The legs attribute is numeric and cannot be used directly with Apriori. Alternatively, you can try to use the Discretize Filter to discretize the legs attribute.
- After deselecting the attributes, use the Apply Filters button to generate a working relation that removes those attributes. Notice how the working relation changes, and has fewer attributes than the base relation.
- First, try using the Apriori algorithm with the default parameters. Record the generated rules.
- Vary the number of rules generated (click on the command that you are running). Try 20, 30, ... Record how many rules you have to generate before generating a rule containing type=mammal.
- Vary the maximum support until a rule containing type=mammal is the top rule generated. Record the maximum support needed. B7. Select one generated rule that was interesting to you. Why was it interesting? What does it mean? Check its confidence and support – are they high enough? B8. Suggest one improvement to the Apriori implementation in WEKA that would have made this data mining lab easier to accomplish.

14. Demonstrate to predict the Numerical Values in the given Data Set is using Regression Methods.

Session-7

15. Demonstrate the classification rule process on the student.arff, employee.arff and labor.arff datasets using the following algorithms:

- a. Logistic Regression
- b. Decision Tree
- c. Naïve Bayes

16. Demonstrate the classification rule process on the student.arff, employee.arff and labor.arff datasets using the following algorithms:

- a. K-Nearest Neighbour
- b. SVM

Session-8

17. Perform the following:

- Demonstrate performing classification on various data sets.
- Load each dataset into WEKA and run ID3, J48 classification algorithm.
- Study the classifier output. Compute entropy values, Kappa statistic.
- Extract if-then rules from the decision tree generated by the classifier.